

Module 08: Introduction to Technology Transfer

Definition of Technology Transfer:

Technology Transfer means sharing skills, knowledge, tools, ways of making things, and facilities between governments, universities, or other organizations. This helps make sure that scientific and technical progress can reach more people and be used to make new products or services.

This can happen:

- From research centers to companies
- Between countries (from developed to developing)
- Inside or between companies (within a company or between companies)

It includes:

- Giving permission to use patents and other intellectual property
- Sharing know-how and technical knowledge
- Giving or selling equipment, tools, and software

Importance of Technology Transfer

1. Helps Innovation Grow

- Connects research centers and companies to help create new products.
- Helps turn research into things people can actually use.

2. Turns Research into Products

- Makes sure discoveries in universities can become useful products or services.
- Helps universities earn money through patents and licenses.

3. Develops Skilled People

- Sharing knowledge also means training people and teaching new skills.
- Local people learn how to use and improve new technology.

4. Builds Self-Reliance

- Reduces the need to depend on other countries for technology.
- Encourages local innovation.

5. Supports Global Partnerships

- Builds international partnerships and good relations.
- Helps countries form long-term technology agreements.

Types of Technology Transfer

Type	Description	Example
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Vertical TT	Transfer from research to production	University → Company
Horizontal TT	Transfer between industries or regions at the same level	Company A → Company B
International TT	Transfer between countries	US → India
Formal TT	Legal transfer involving patents, licensing, etc.	Patent licensing agreements
Informal TT	Transfer through training, publications, or movement of personnel	Workshops, research papers

Benefits of Technology Transfer

- Helps new products reach the market faster
- Improves productivity and makes companies more competitive
- Creates jobs by building new industries
- Solves real-life problems using science and research
- Brings money to universities and governments

Example of Technology Transfer

Example 1: COVID-19 Vaccine

- Universities like Oxford University made a COVID-19 vaccine.
- The technology was given to AstraZeneca, a drug company.
- AstraZeneca made and sold the vaccine worldwide, even to poor countries at a low cost.

Example 2: Indian Space Research Organization (ISRO)

- ISRO shares satellite and space technology with Indian companies to make parts.
- It has also shared spin-off tech like thermal insulation, GPS, and food storage methods for everyday use.

Example 3: Green Revolution in India

- High-yield crop technology and irrigation methods were shared from rich countries to India.
- This helped India grow more food and become self-sufficient in farming.

Historical Evolution of Technology Transfer

Technology Transfer means moving knowledge, ideas, or methods from one place to another so they can be used and turned into products. It has changed over time due to science, industry, and world trends.

1. Pre-Industrial Era (Before 18th Century)

What it was like:

- Sharing of technology was informal.
- Knowledge was passed by word of mouth or through apprenticeships.
- Inventions were local and made by trial and error.

Example:

- Farming techniques like irrigation and crop rotation were shared among ancient civilizations like Mesopotamia, Egypt, and the Indus Valley.
- Papermaking moved from China to the Islamic world, then to Europe.

2. Industrial Revolution (18th to 19th Century)

What it was like:

- Machines and science increased the need for organized knowledge sharing.
- Governments and companies began protecting inventions with patents.
- Machines and know-how started moving across countries.

Example:

- Textile machines invented in Britain were copied and taken to the U.S. by Samuel Slater in the late 1700s (he's called the "Father of the American Industrial Revolution").
- Railway technology spread from Britain to Europe, the US, and its colonies.

3. Early 20th Century (1900–1945)

What it was like:

- Big companies built their own research labs (like General Electric, DuPont).
- Universities started doing more applied research.
- Governments spent money on tech for the military.

Example:

- Aircraft technology advanced during the World Wars and was shared between allies.
- Synthetic rubber was developed during World War II when access to natural rubber was cut off. This tech was shared between countries and industries.

4. Post-War Era & Cold War Period (1945–1990)

What it was like:

- Many organized R&D centers were created.
- Countries made official agreements for sharing tech.

- Multinational companies (MNCs) grew.
- Universities began to license their technologies.

Example:

- The Marshall Plan (1948): The US helped rebuild Europe and Japan by giving industrial technology.
- NASA's space tech was later used for everyday things like memory foam and scratch-proof lenses.

5. Modern Globalization Era (1990–2010)

What it was like:

- Huge growth in digital technology.
- Shift to knowledge-based economies.
- More international teamwork on R&D.
- Rise of tech parks and startup incubators.

Example:

- Silicon Valley's system of working together (university, industry, government) spread to other countries.
- Open-source software like Linux made global sharing of technology possible.

6. Contemporary Era (2010–Present)

What it is like:

- Focus on green, sustainable, and AI-based technologies.
- Startups and investors play a big role in making new tech.
- Intellectual property rules became more important and complex.
- More technology sharing between developing countries (South-South).

Example:

- COVID-19 vaccine tech: mRNA vaccine technology was shared between countries for quick mass production.
- India's space tech: ISRO's satellite tech is now used by startups for farming, transport, and more.

Role of Universities, Research Institutions, and Industries in Technology Transfer

1. Role of Universities

► Definition:

Universities are places for higher education where people learn, do research, and build skills.

► Roles:

• Education and Talent Development:

- o Teach students to become skilled graduates and postgraduates.
- o Build critical thinking and problem-solving skills.

• Research and Innovation:

- o Do basic and applied research.
- o Create new ideas, technologies, and solutions.

• Sharing Knowledge:

- o Publish research in journals and present at conferences.
- o Hold workshops, seminars, and meetings.

• Working with Industry:

- o Work with companies on research projects and student internships.

► Example:

- IITs (Indian Institutes of Technology) in India work with ISRO and DRDO on space and defense projects.
- MIT (USA) works with Google and IBM to research Artificial Intelligence.

2. Role of Research Institutions

► Definition:

Research institutions are centers that focus only on scientific and industrial research, both basic and advanced.

► Roles:

• Focused Research:

- o Work on specific scientific, technical, or social problems.

• Support for Policies:

- o Give advice to the government using scientific facts and studies.

• Technology Development:

- o Create new technologies for use in public and private sectors.

• Building Capacity:

- o Provide training and scholarships to researchers and professionals.

► **Example:**

- CSIR (Council of Scientific and Industrial Research) in India made a cheap water purifier now used in villages.
- NASA (USA) works with universities and companies to explore space and make new space technology.

3. Role of Industries

► **Definition:**

Industries are businesses that make and sell products or services.

► **Roles:**

- **Turning Technology into Products:**
 - Take research and turn it into real products or services.
- **Funding and Investment:**
 - Pay for university research and invest money in new ideas.
- **Creating Jobs:**
 - Hire skilled people from universities and research centres.
- **Giving Feedback and Identifying Needs:**
 - Guide research by sharing what the market and customers need.

► **Example:**

- Tata Consultancy Services (TCS) works with universities to improve software research.
- Tesla works with battery labs to make better energy storage systems.

Challenges and Barriers in Technology Transfer

Technology transfer means moving new ideas or inventions from research centres or one group to another so they can be developed, sold, or used. Although this helps innovation and economic growth, many problems can make it difficult.

1. Intellectual Property (IP) Issues

Challenge:

It can be unclear who owns the rights to an idea or invention, which can delay technology transfer.

Example:

A university researcher makes new software with money from both the government and a company. Deciding who owns it and how to license it is complicated.

2. Lack of Communication and Collaboration

Challenge:

When researchers and companies don't communicate well, the technology might not fit real-world needs.

Example:

A lab makes a new medical device but doesn't talk enough with hospitals, so the device doesn't work well in real hospitals.

3. Cultural and Organizational Differences

Challenge:

Universities and companies have different goals and ways of working, which can create problems.

Example:

Universities want to publish knowledge, but companies want profit and secrecy, which can cause conflict in projects.

4. Inadequate Funding and Resources

Challenge:

Turning research into a product can cost a lot, and it's hard to find investors willing to take that risk.

Example:

A startup has an idea for new green energy, but can't find investors because the cost to make it real is too high at first.

5. Technological Complexity and Maturity

Challenge:

Sometimes the technology is not ready for the market and needs more research and testing.

Example:

A new nanotechnology material for planes works only in the lab and needs many more years of work before it can be used.

6. Regulatory and Legal Barriers

Challenge:

Laws about safety, quality, environment, or ethics can delay or stop new technologies.

Example:

A new gene-editing tool is stuck in approval for years because of strict rules and ethical checks.

7. Lack of Skilled Human Resources

Challenge:

Some companies don't have people with the right skills to use new technology.

Example:

A small factory buys a license for an advanced AI tool, but doesn't have data scientists to use it.

8. Market Uncertainty and Risk Aversion

Challenge:

Companies may not want to risk using new, untested technologies.

Example:

A new cybersecurity tool is better, but companies stick with old tools because they are afraid of bugs and lack of support.

9. Geopolitical and Economic Barriers

Challenge:

Sharing technology between countries can be stopped by trade rules, sanctions, or economic issues.

Example:

A US semiconductor company can't share technology with a Chinese company because of export restrictions.

10. Poor Infrastructure and Ecosystem Support

Challenge:

In some developing areas, poor infrastructure makes it hard to use new technology.

Example:

A new solar energy system can't be used in a rural area because there's no place to fix or maintain it.